Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

System mass

Meaning in my words How would you explain system mass to a friend?	**How can you tell?** If a cart and hanging mass accelerate together, which masses belong in the total?
Example or sketch A loaded cart is 0.8 kg and its loaded hanger is 0.2 kg. What mass accelerates?	**Non-example and why** Why leave the fixed track out of the moving mass?

Control

Meaning in my words How would you explain control to a friend?	**How can you tell?** You test different cart masses. Why use the same hanging weight each time?
Example or sketch Name two things you would set the same before each release.	**Non-example and why** A partner lists cart mass as a control while adding mass each trial. Where does it belong?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Inverse relationship

Meaning in my words How would you explain inverse relationship to a friend?	**How can you tell?** Keep the net force fixed. If the moving mass doubles, what happens to acceleration?
Example or sketch At 4 N, compare the accelerations of carts of 1 kg and 2 kg.	**Non-example and why** If both mass and net force double, must acceleration halve?

Slope

Meaning in my words How would you explain slope to a friend?	**How can you tell?** At the same net force, which graph should be straight: a against M or a against 1/M?
Example or sketch Acceleration rises by 2 m/s² when 1/M rises by 0.5 per kg. What is the slope?	**Non-example and why** Someone says this slope is the mass. How can you check?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch a versus 1/M. Label the axes and units, draw the rising line, and name the quantity represented by its slope.



My marks mean

Same push. Different mass.

11

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Compare velocity changes when masses differ.

CHOOSE ONE JOB TOGETHER

TEST ENGINEER ☐

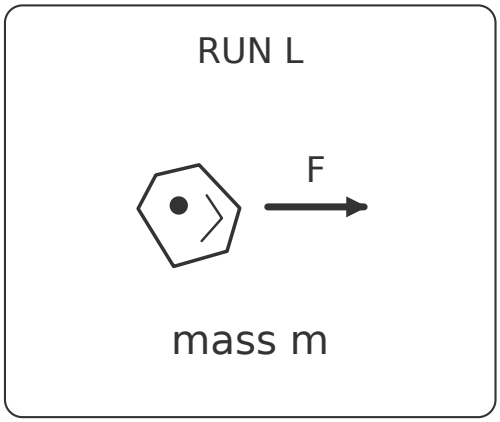
Compare the two test runs. Tell the crew which body changes velocity more and use the masses to explain why.

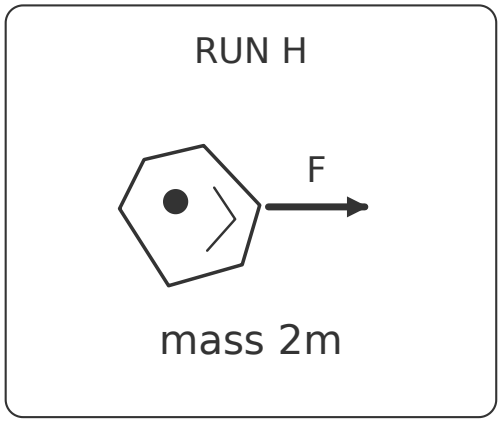
LAB CORRESPONDENT ☐

Explain why equal pushes can give different results. Have your partner check what stayed the same in both tests.

Point to a clue. Keep your first marks; add the update using the color key.

Two asteroid models get the same net force for the same time. Model H has twice the mass of L.





01 OUR FIRST CALL

Which body changes velocity more? Explain your first call from the test card.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

What changed in the test? Give the crew a new comparison.

Switch speakers. Check both runs using mass times acceleration. Do your results match the forces on the test card?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks